IAC TECHNOLOGY DIVISION

TITLE:

MUSOLSONG COORDINATOR SOFTWARE SYSTEM - USER MANUAL

PROJECT / DESTINATION:

MUSOL - ESPECTRO POLARÍMETRO SOLAR

Document Code: DS/TN-MUS/0012v1.0

Issue: 1.0
Date: 03-03-2026
External code: N/A
PBS code: N/A
Work Package: N/A
Status: N/A

INSTITUTO DE ASTROFISICA DE CANARIAS
38200 La Laguna (Tenerife) - ESPAÑA - Phone (922)605200 - Fax (922)605210

AUTHORS LIST

Name	Institution	Function	Date	Signature
Heidy Moreno Arce	IAC		03-03-2026	

APPROVAL CONTROL

	Name	Institution	Function	Date	Signature
Revised				03-03-2026	
Approved				03-03-2026	
Authorized				03-03-2026	

DOCUMENT CHANGES RECORD

Issue	Date	Change Description
1.0	03-03-2026	Initial Issue

SUMMARY

This document describes the use of the MUSOLSONG controller software. A software system designed to coordinate calibration and observation procedures between the MUSOL polarimeter and SONG spectrograph.

LIST OF ABBREVIATIONS

Abbreviation	Description
AD	Applicable Document
IAC	Instituto de Astrofísica de Canarias
RD	Referenced Document
TBC	To Be Confirmed
TBD	To Be Defined

TABLE OF CONTENTS

AUTHORS LIST **2**

APPROVAL CONTROL **2**

DOCUMENT CHANGES RECORD **2**

SUMMARY **3**

THIS DOCUMENT DESCRIBES THE USE OF THE MUSOLSONG CONTROLLER SOFTWARE. A SOFTWARE SYSTEM DESIGNED TO COORDINATE CALIBRATION AND OBSERVATION PROCEDURES BETWEEN THE MUSOL POLARIMETER AND SONG SPECTROGRAPH **3**

LIST OF ABBREVIATIONS **4**

1. INTRODUCTION **6**

1.1. BACKGROUND 6

2. OVERVIEW **6**

2.1. SYSTEM STRUCTURE 6

2.2. SYSTEM REQUIREMENTS 6

2.3. INSTALLATION 6

2.4. SYSTEM CONFIGURATION FILE: 7

2.5. GETTING STARTED$ 7

2.5.1. Running on GUI mode 7

2.5.2. Running on CLI mode 7

2.5.3. Running the app examples 8

2.5.4. Initial System Check 9

2.5.5. Status Indicators on the GUI 9

2.5.6. Main GUI Overview 9

2.5.7. Configuration Details 12

2.5.8. Troubleshooting & Support 12

2.6. KEY FEATURES 12

2.6.1. Sequence Management 12

2.6.2. Automated Processing 13

2.6.3. Manual Control 13

2.6.5. Command Line Interface (CLI) 13

2.6.6. System Configuration (systemConfig.yaml) 13

2.6.7. Sequence Files (YAML) 13

2.6.8. Log Files 13

1. INTRODUCTION

1.1. Background

MUSOLSONG is part of a research project aimed at upgrading the Solar-SONG instrument, which operates on the Hertzsprung-SONG telescope at the Teide Observatory. The main objective of this project is to study the Sun's global and cyclical magnetism, complemented by investigations of its interior to better understand the origin of our star's cyclical activity.

The project involves the design, manufacture, and integration of the MUSOL polarimeter, its coupling with the SONG spectrograph, and the provision of all the necessary infrastructure to ensure the proper operation of the polarimeter.

2. OVERVIEW

MUSOLSONG is a software project containing two software applications: The musolsong-controller and the musolsong-yamleditor.

The MUSOLSONG control system coordinates operations between the MUSOL polarimeter and the SONG spectrograph using a modular Python-based application with GUI (Graphical User Interface) and a CLI (Command Line Interface). Operational sequences are defined via YAML files and executed for calibration and observation procedures enabling automated control of spectro-polarimeter operations.

2.1. System Structure

- **Controller (GUI/CLI)**: Centralized application for operations management
- **YAML Editor**: Application used to create/edit configuration files for the controller

2.2. System Requirements

Operating System: The application was developed on Linux OS. However, it should be compatible with other systems (Windows or mac Os) supporting Python and PyQt6

Required Python Version:

Python 3.6 or higher

Required Packages:

PyQt6 >=6.91,

PyYAML>=5.4.1,

musolLib (uses **pyads** 3.5.2 to communicate with Twin-CAT PLCs via ADS).

External Hardware: Beckhoff PLC, MUSOL polarimeter, SONG spectrograph

2.3. Installation

pip install musolsong-2025.0.1.0-py3-none-any.whl

2.4. System configuration file:

A **`systemConfig.yaml`** file is included with the **musolsong** package. This file contains information about connection parameters for hardware components (PLC and SONG IPs and ports) and configuration settings for system logging behavior. Note that the current IP and port values are those used for testing in the IAC lab. Values to be used at the observatory are **TBD**

Location:

$HOME/.local/lib/python3.10/site-packages/musolsong/controller/etc/systemConfig.yaml

Default file:

```
plc_MUSOL:
  ams_net_id: "5.110.208.2.1.1"  # PLC AMS Net ID of the polarimeter
  logs_enabled: true
  logs_path: "/$HOME/logs"     # Directory for Musol library logs

spectrograph_SONG:
  host: "127.0.0.1"   # IP address of the SONG server
  port: 8050         # Port number of the SONG server
  timeout: 30       # Timeout in seconds

MUSOLSONG_system:
  logs_enabled: true              # Set to true or false
  logs_path: "NONE"            # Logs directory path, if do not want a log file,  state NONE
  logs_level: "DEBUG"        # Log level
  logs_console_enabled: true   # Set to true or false
```

2.5. Getting Started$

The application can run in either GUI or CLI mode, and in either real or simulation mode.

2.5.1. Running on GUI mode

To run the **GUI** version in **real mode**, simply start the application with:

```
musolsong-controller
```

To run in **simulation mode**, both the musolLib and the SONG server , first launch the SONG simulator software.

The SONG server simulator must be started in a separate terminal using:

```
musolsong-songsimulator
```

Once the simulator is up and running, start the controller by typing:

```
USE_MUSOLLIB_SIMULATOR=1 musolsong-controller
```

2.5.2. Running on CLI mode

The CLI interface can be used to process a sequence modulation file or to simply validate a file if the --validate-only flag is selected. Traces are shown on the standard output if verbose mode is enabled.

Usage:

musolsong-controller [--sequence-yaml SEQUENCE_YAML] [--project-name PROJECT_NAME] [--project-number PROJECT_NUMBER] [--validate-only] [--verbose] [--help]

Options:

General

Option	Alias	Description
--help	-h	Show this help message and exit

CLI Mode Arguments

Option	Alias	Description
--sequence-yaml SEQUENCE_YAML		Path to sequence YAML modulations file (enables CLI mode) |
--project-name PROJECT_NAME	-n PROJECT_NAME	Project name (required for CLI mode)
--project-number PROJECT_NUMBER	-p PROJECT_NUMBER	Project number (required for CLI mode)
-validate-only		Only validate the YAML file without processing
--verbose	-v	Enable verbose output

2.5.3. Running the app examples

GUI Mode

musolsong-controller

CLI Mode

musolsong-controller --sequence-yaml config.yaml --project-name "My Project" --

project-number 123

CLI Validate Only

musolsong-controller --sequence-yaml config.yaml -n "Test" -p 456 --validate-only

2.5.4. Initial System Check

Upon startup, the system will:

- Load configuration from `systemConfig.yaml`
- Initialize the system logger
- Attempt connection to MUSOL PLC
- Attempt connection to SONG spectrograph
- Display connection status in the status bar (GUI mode)

2.5.5. Status Indicators on the GUI

- **Green Status Bar**: Both PLC and SONG server connected and ready
- **Red Status Bar**: One or both systems not available
- Only the Engineering tab is available if connections of one of the systems fail

2.5.6. Main GUI Overview

1.- Menu Options

- **File → Load sequence YAML file**: Load predefined observation/calibration sequences
- **File → Exit**: Close the application

2.- Project Information

- Enter project number and name at startup; these are used by the SONG server for FITS headers.
    - **Project Number**: Numerical identifier (default: 0) as defined in the soda.phys.au.dk by the SONG Sci. team
    - **Project Name**: Descriptive name for the observation session as defined in the soda.phys.au.dk by the SONG Sci. Team
- Select between Operations or Engineering tabs.

3.-Operations Tab Workflow

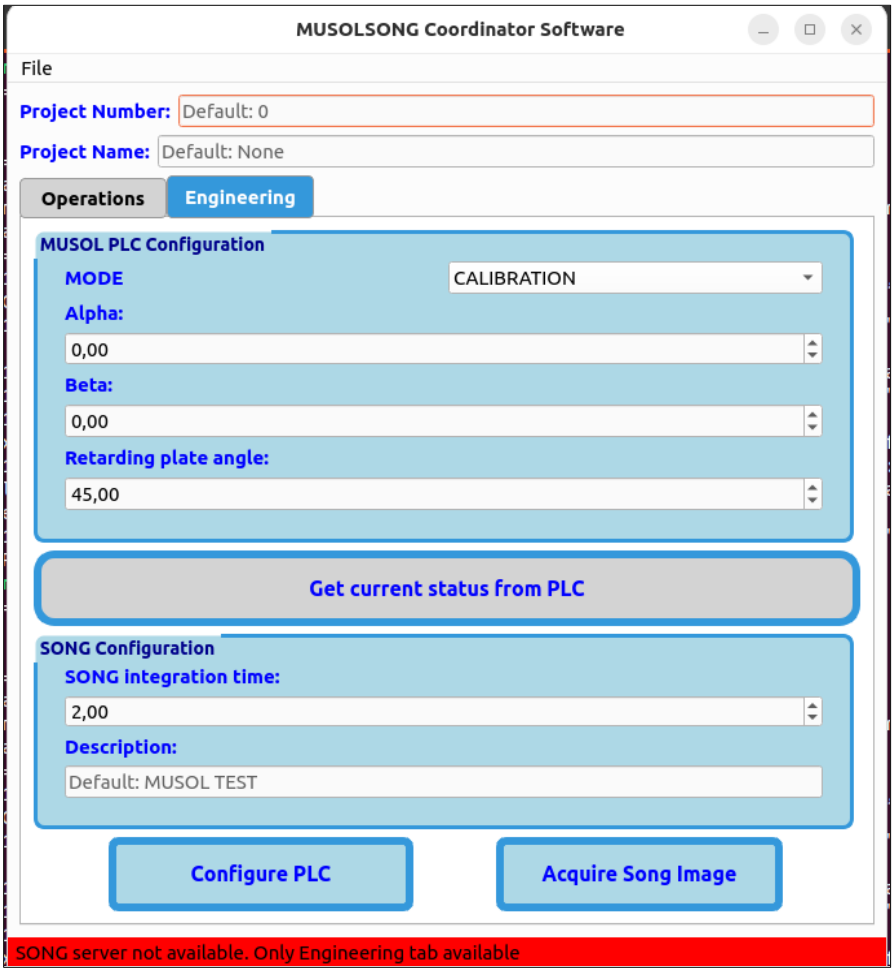
Used for automated sequence processing:

1. **Load a YAML** sequence file (menu File > Load sequence YAML file). File is validated and populates the modulations area, displaying mode (CALIBRATION/OBSERVATION). Should note that it is recommended to create YAML files by using the ***musolsong-yamleditor*** application. If files are not created with the ***yamleditor*** app, validate the YAML

syntax using standard validators before loading.

2. **Modulations Table**: Displays the loaded observation/calibration sequence steps with Alpha, Beta angles, integration time, and descriptions
3. **If CALIBRATION:** Shows incremental angle per cycle; the system computes the number of cycles and retarder theta offsets are managed per cycle.
4. Press **"START Processing"** to execute the loaded sequence. Each modulation is sent to the polarimeter, along with a request to acquire an image to the SONG server. The name of the created FITS image is displayed in the status bar at the bottom.
5. "Abort" button available at any time.
6. Real time status updates and progress tracking are displayed in the progress bar, current status area, and system status updates bar

4.- Engineering Tab Workflow



Use this tab for manual or test operations.

Configures the polarimeter and sends acquisition commands individually.

- **MUSOL PLC Configuration**:

  1.-Enter:

  - Operational mode selection (CALIBRATION/OBSERVATION)

- Alpha and Beta angle controls
- Retarding plate angle configuration (CALIBRATION mode only)

2.-Press “Configure PLC” button to send the configuration to PLC

- **PLC Current Status**:
  - Press “Get current status from PLC” button to request the current operational mode and motor positions.
  - Received status information is shown on the current Status Bar

- **SONG Configuration**:

Enter:

- Integration time settings
- Acquisition description
- Press “Acquire SONG Image” button to send image acquisition to SONG spectrograph server

2.5.7. Configuration Details

- Required configuration file: systemConfig.yaml in $HOME/.local/lib/python3.10/site-packages/musolsong/controller/etc/systemConfig.yaml
- Logging setup: Enabled/disabled, directory path, log level, console display.
- Sequence YAML files: Must specify correct mode, modulations, and relevant parameters.

2.5.8. Troubleshooting & Support

Common Issues

Connection Failures:

- Verify network connectivity to MUSOL PLC and SONG server
- Check AMS Net ID and IP/port settings in configuration
- Ensure hardware systems are powered on and accessible

YAML File Loading Errors:

- Validate YAML syntax using standard validators
- Check sequence file against expected schema

Processing Failures:

- Verify both systems are connected before starting sequences
- Check log files for detailed error information

2.6. Key Features

2.6.1. Sequence Management

- Load YAML files defining observation/calibration sequences
- Validate sequence configurations before execution

- Visual representation of modulation steps

2.6.2. Automated Processing

- Execute complete observation sequences automatically
- Real-time progress monitoring
- Safe abort capability during execution

2.6.3. Manual Control

- Direct control of MUSOL polarimeter parameters
- Manual SONG spectrograph image acquisition
- Engineering controls for testing and troubleshooting

2.6.4. Simulation Mode

- Built-in simulators for MUSOL library and SONG server
- Test functionality without hardware connections
- Development and training capabilities

2.6.5. Command Line Interface (CLI)

- Automation and Integration: Easily integrated into **scripts.**(e.g., batch processing with shell scripts).
- Remote accessibility: Can be used over **SSH** or in headless environments — no GUI required.

2.6.6. System Configuration (`systemConfig.yaml`)

- Connection parameters for hardware components (PLC and SONG IPs and Ports)
- Logging settings and paths
- System behavior configurations

2.6.7. Sequence Files (YAML)

Define observation sequences with:

- Operational mode (observation/calibration)
- Modulation parameters (alpha, beta angles)
- Integration times
- Descriptions for each step
- Calibration-specific settings (offset angles, cycles)

2.6.8. Log Files

System generates detailed logs with timestamps for debugging purposes. These files are generated on the logs_path directory specified in the system configuration file if requested. There is also detail operation logs of the PLC musol library.